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Flash memory device with three-dimensional half structure and methods for forming the same

Abstract

A flash memory device and method of making the same are disclosed. The flash memory device is located on a substrate and includes a floating gate electrode, a tunnel dielectric layer located between the substrate and the floating gate electrode, a smaller length control gate electrode and a control gate dielectric layer located between the floating gate electrode and the smaller length control gate electrode. The length of a major axis of the smaller length control gate electrode is less than a length of a major axis of the floating gate electrode.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation application of U.S. application Ser. No. 17/868,192 entitled “Flash Memory Device with Three-Dimensional Half Flash Structure and Methods for Forming the Same,” filed on Jul. 19, 2022, which is a divisional application of U.S. application Ser. No. 17/191,334 entitled “Flash Memory Device with Three-Dimensional Half Flash Structure and Methods for Forming the Same,” filed on Mar. 3, 2021 now patented as U.S. Pat. No. 11,658,248, the entire contents of both of which are incorporated herein by reference for all purposes.

BACKGROUND

(1) To program flash memory, a voltage may be applied to the control gate resulting in charge accumulation in the floating gate. The charge accumulation in the floating gate may result in induced various capacitances. For example, induced capacitances may form between the control gate and the floating gate ($C_{\text{sub.ONO}}$), between the drain and the floating gate ($C_{\text{sub.D}}$), between the source and the floating gate ($C_{\text{sub.S}}$) and between the floating gate and the bulk semiconductor as measured in the channel ($C_{\text{sub.B}}$). The coupling ratio of a flash memory cell may be defined as the ratio of the capacitance between the control gate and the floating gate $C_{\text{sub.ONO}}$ to the sum of the capacitances ($C_{\text{sub.ONO}}+C_{\text{sub.D}}+C_{\text{sub.S}}+C_{\text{sub.B}}$). Flash memory cells with a higher coupling ratio may achieve faster programming times than flash memory cells with a lower coupling ratio.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a vertical cross-sectional view illustrating a step of forming a continuous tunnel dielectric layer, a continuous floating gate layer, a continuous control gate dielectric layer, a continuous control gate layer and photoresist layer on a substrate in a method for fabricating a semiconductor device in accordance with some embodiments.
- (3) FIG. 2 is a vertical cross-sectional view illustrating a step of patterning the photoresist layer in a method for fabricating a semiconductor device in accordance with some embodiments.
- (4) FIG. 3 is a vertical cross-sectional view illustrating a step of patterning the continuous tunnel dielectric layer, the continuous floating gate layer, the continuous control gate dielectric layer and the continuous control gate layer with the patterned photoresist.
- (5) FIG. 4 is a vertical cross-sectional view illustrating a step of ion implanting the substrate to form active extension regions in the substrate in a method for fabricating a semiconductor device in accordance with some embodiments.
- (6) FIG. 5 is a vertical cross-sectional view illustrating a step of depositing a photoresist in a method for fabricating a semiconductor device in accordance with some embodiments.
- (7) FIG. 6 is a vertical cross-sectional view illustrating a step of patterning a photoresist in a method for fabricating a semiconductor device in accordance with some embodiments.
- (8) FIG. 7 is a vertical cross-section view illustrating a step of further etching the control gate electrode using the patterned photoresist as a mask.
- (9) FIG. 8 is a vertical cross-sectional view illustrating a step of forming a continuous sidewall spacer layer in a method for fabricating a semiconductor device in accordance with some

embodiments.

(10) FIG. 9 is a vertical cross-sectional view illustrating a step of forming sidewall spacers by etching the continuous sidewall spacer layer in a method for fabricating a semiconductor device in accordance with some embodiments.

(11) FIG. 10 is a vertical cross-sectional view illustrating a step of ion implanting the substrate to form deep source and drain regions a method for fabricating a semiconductor device in accordance with some embodiments.

(12) FIG. 11 is a vertical cross-sectional view illustrating a step of siliciding the deep source and drain regions and the gate electrode in a method for fabricating a semiconductor device in accordance with some embodiments.

(13) FIG. 12 is a vertical cross-sectional view illustrating a step of forming an interlevel dielectric layer and source/drain electrodes in a method for fabricating a semiconductor device in accordance with some embodiments.

(14) FIG. 13 is a top-down view of an array of flash memory cells in accordance with some embodiments.

(15) FIG. 14 is a flowchart illustrating a method of making a semiconductor memory device in accordance with some embodiments.

DETAILED DESCRIPTION

(16) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(17) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(18) Generally, the structures and methods of the present disclosure can be used to form flash memory devices in which at least some of the flash memory devices have a control gate electrode that is shorter in length than the length of the corresponding floating gate electrode. Further, at least some of the flash memory devices have at least one electrical contact to the gate dielectric layer in addition to the electrical contacts to the source region, drain region and control gate electrode. The floating gate electrode can be programmed and erased by applying a voltage to the electrical contact connected to the gate dielectric layer. By using the electrical contact to the gate dielectric layer for programming and erasing rather than using the control gate electrode, the resistance $R_{\text{sub.poly}}$ associated with the polysilicon control gate electrode can be avoided. The elimination of the resistance $R_{\text{sub.poly}}$ associated with the polysilicon control gate electrode may result in a better coupling ratio which results in faster programming and erasing of the flash memory device.

(19) In some embodiments, more than one electrical contact may be provided to the gate dielectric layer. The use of more than one electrical contact to the gate dielectric layer may result in an increased coupling area. The increased coupling area may in turn result in a higher coupling ratio.

The coupling ratio is defined by equation 1 below:

$$(20) \quad c_{cg} = \frac{C_{ONO}}{(C_D + C_S + C_B + C_{ONO})} \quad \text{Equation 1}$$

(21) Where a $\alpha_{sub.cg}$ is the coupling ratio, $C_{sub.ONO}$ is the capacitance between the control gate and the floating gate, $C_{sub.D}$ is the capacitance between the drain and the floating gate, $C_{sub.S}$ is the capacitance between the source and the floating gate, and $C_{sub.B}$ is the capacitance between the floating gate and the bulk semiconductor as measured in the channel. As discussed above, the increase in the coupling ratio may result in faster programming and erasing of the flash memory device.

(22) In conventional flash memory, each memory cell resembles a standard metal-oxide-semiconductor field-effect transistor (MOSFET) except that the transistor has two gates instead of one. The memory cells can be seen as an electrical switch in which current flows between two active regions (source and drain) and is controlled by a floating gate (FG) and a control gate (CG). The CG is similar to the gate in other MOS transistors, but below this, there is the FG insulated all around by an oxide layer. The FG is interposed between the CG and the MOSFET channel. Because the FG is electrically isolated by its insulating layer, electrons placed on it are trapped. When the FG is charged with electrons, this charge screens the electrical field from the CG, thus, increasing the threshold voltage ($V_{sub.T1}$) of the cell. Thus, a higher voltage ($V_{sub.T2}$) must be applied to the CG to make the MOSFET channel conductive. In order to read a value from the transistor, an intermediate voltage between the threshold voltages ($V_{sub.T1}$ $V_{sub.T2}$) is applied to the CG. If the channel conducts at this intermediate voltage, the FG must be uncharged (if it was charged, we would not get conduction because the intermediate voltage is less than $V_{sub.T2}$), and hence, a logical “1” is stored in the gate. If the channel does not conduct at the intermediate voltage, it indicates that the FG is charged, and hence, a logical “0” is stored in the gate. The presence of a logical “0” or “1” is sensed by determining whether there is current flowing through the transistor when the intermediate voltage is asserted on the CG. In a multi-level cell device, which stores more than one bit per cell, the amount of current flow is sensed (rather than simply its presence or absence), in order to determine more precisely the level of charge on the FG.

(23) In various embodiments, the status of the memory device (i.e. logical “1” or “0”) can be determined with the contact to the control gate electrode as in other flash memory devices.

(24) Referring to FIG. 1, a continuous tunnel dielectric layer **104L**, a continuous floating gate layer **106L**, a continuous control gate dielectric layer **114L**, a continuous control gate layer **116L** and a continuous photoresist layer **118L** may be formed on a semiconductor substrate **102**. The semiconductor substrate **102** may be made of any suitable material, such as silicon, silicon on insulator (SOI) or silicon on sapphire (SOS). The semiconductor substrate **102** may include electrical dopants (such as p-type dopants or n-type dopants) at an atomic concentration less than $1.0 \times 10^{14}/\text{cm}^3$ in order to provide low electrical conductivity and to minimize eddy current that may be induced by inductive coupling with high frequency electrical signals from, to, or between semiconductor dies or redistribution wiring interconnects to be subsequently placed in proximity.

(25) The continuous tunnel dielectric layer **104L** may be made of any suitable material, such as an oxide or nitride, such as silicon oxide. The continuous floating gate layer **106L** may be made of any suitable material, such as poly silicon. The continuous control gate layer **116L** may be made of a metal, such as tungsten, nickel, aluminum or alloys thereof or the continuous control gate layer **116L** may be made of be made of poly silicon.

(26) The continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L** and the continuous control gate layer **116L** may be each deposited by any suitable method, such as chemical vapor deposition (CVD), organometallic chemical vapor deposition (OMCVD), plasma enhanced chemical vapor deposition (PECVD), low pressure chemical vapor deposition (LPCVD) or atomic layer deposition (ALD). The continuous

photoresist layer **118L** may be a positive or negative photoresist.

(27) In various embodiments, the continuous control gate dielectric layer **114L** may include a continuous first oxide layer **108L**, a continuous nitride layer **110L** and continuous second oxide layer **112L**. This layered structure may be referred to as a oxide-nitride-oxide (ONO) sandwich layer. A continuous first oxide layer **108L** may be deposited over the continuous floating gate layer **106L**. The continuous first oxide layer **108L** may be made of any suitable oxide, such as SiO₂, HfO₂, Al₂O₃, Ta₂O₅, ZrO₂, TiO₂. Next, a continuous nitride layer **110L**, such as silicon nitride, may be deposited over the continuous first oxide layer **108L**.

(28) A continuous second oxide layer **112L** may be deposited over the continuous nitride layer **110L**. In this manner, a continuous oxide/nitride/oxide control gate dielectric layer **114L** may be formed over the continuous floating gate layer **106L**. The continuous second oxide layer **112L** may be made of the same material as the continuous first oxide layer **108L** or may be made of a different material. The continuous first oxide layer **108L** may have a thickness in the range of 20 Å-200 Å, such as 50 Å-150 Å, although lesser or greater thicknesses are within the contemplated scope of disclosure. The continuous nitride layer **110L** may have a thickness in the range of 20 Å-250 Å, such as 50 Å-200 Å, although lesser or greater thicknesses are within the contemplated scope of disclosure. The continuous second oxide layer **112L** may have thickness in the range of 100 Å-500 Å, such as 200 Å-400 Å, although lesser or greater thicknesses are within the contemplated scope of disclosure. In an embodiment, a thickness of the continuous second oxide layer **112L** may be 5-10% greater than a thickness of the continuous first oxide layer **108L**.

(29) Referring FIG. 2, the continuous photoresist layer **118L** may be exposed to radiation and patterned. As illustrated in FIG. 2, the continuous photoresist layer **118L** may be patterned to form a patterned photoresist layer **118** such that a central portion of the continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L**, and the continuous control gate layer **116L** are masked by the patterned photoresist layer **118**.

(30) Referring FIG. 3, the patterned photoresist layer **118** may then be used to mask and pattern the continuous control gate layer **116L**, the continuous control gate dielectric layer **114L** (including continuous second oxide layer **112L**, continuous nitride layer **110L**, and the continuous first oxide layer **108L**), the continuous floating gate layer **106L**, and the continuous tunnel dielectric layer **104L**. The result may be a stack **119** of patterned layers **104**, **106**, **114**, **116** that includes a patterned tunnel dielectric layer **104**, a floating gate electrode **106**, a patterned control gate dielectric layer **114** and a control gate electrode **116**. In various embodiments, the patterned control gate dielectric layer **114** may include an ONO sandwich layer comprising a patterned first oxide layer **108**, a patterned nitride layer **110** and a patterned second oxide layer **112**. The continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L**, and the continuous control gate layer **116L** may be patterned by one or more wet etching processes, one or more dry etching processes or a combination of wet and dry etching processes.

(31) Referring to FIG. 4, the semiconductor substrate **102** may be subject to a first ion implantation step **120** to form active extension regions **122**, **124** (i.e. source/drain extension regions) in the semiconductor substrate **102**. The stack **119** of patterned layers may serve to mask the portion of the semiconductor substrate **102** between the active extension regions **122**, **124**. Therefore, active extension regions **122**, **124** may be self-aligned to the floating gate electrode **106** and the control gate electrode **116**. The active extension regions **122**, **124** may be doped either p-type or n-type as desired. Example n-type dopants include, but are not limited to, antimony, arsenic and phosphorous. Example p-type dopants include, but are not limited to boron, aluminum and gallium. The active extension regions **122**, **124** may have an implanted dopant concentration of 1×10^{20} to 5×10^{21} , such as 2×10^{20} to 2×10^{21} , although lesser or greater dopant concentrations are within the contemplated scope of disclosure. The region under the stack **119** of patterned layers located between active extension regions **122**, **124** constitutes the channel region **125**. The patterned photoresist layer **118** may be removed by any suitable process, such as an

ashing process. Alternatively, the patterned photoresist layer **118** may be removed prior to the first ion implantation step **120** occurs.

(32) Referring to FIGS. **5** and **6**, a second photoresist layer **118a** may be deposited over the semiconductor substrate **102** and the stack **119** of patterned layers **104**, **106**, **114**, **116**. The second photoresist layer **118a** may then be patterned to expose a portion of the stack **119** of patterned layers **104**, **106**, **114**, **116**. The second photoresist layer **118a** may be made of the same material as the material of the patterned photoresist layer **118** or may be made of a different material.

(33) Referring to FIG. **7**, the exposed portion of the control gate electrode **116** may be etched to form a smaller length control gate electrode **116a**. That is, the control gate electrode **116** may be etched such that the length $L_{sub.CG}$ of the smaller length control gate electrode **116a** along its major axis, is shorter than the length $L_{sub.FG}$ of the underlying floating gate electrode **106** along a major axis of the floating gate electrode **106**. The control gate electrode **116** may be either wet etched or dry etched to form the smaller length control gate electrode **116a**. The underlying patterned control gate dielectric layer **114** may serve as an etch stop. The remaining second photoresist layer **118a** may be removed by any suitable process, such as an ashing process.

(34) Referring to FIG. **8**, a continuous sidewall spacer layer **126L** may be conformally deposited over the surface of the semiconductor substrate **102** and the stack **119** of patterned layers **104**, **106**, **114**, **116a**. The continuous sidewall spacer layer **126L** may be made of any suitable dielectric material, such as SiO_{2sub} or Si_3N_{4sub} . The continuous sidewall spacer layer **126L** may be deposited by any suitable method, such as CVD, OMCVD, PECVD, LPCVD or ALD.

(35) Referring to FIG. **9**, the continuous sidewall spacer layer **126L** may be anisotropically etched to form a first sidewall spacer **126a** located over sidewalls of the patterned tunnel dielectric layer **104**, the floating gate electrode **106** and the patterned control gate dielectric layer **114**, a second sidewall spacer **126b** located over sidewalls of the smaller length control gate electrode **116a** and a third sidewall spacer **126c** located over sidewalls of the patterned tunnel dielectric layer **104**, the floating gate electrode **106**, the patterned control gate dielectric layer **114** and the smaller length control gate electrode **116a**. The anisotropic etching process removes portions of the continuous sidewall spacer layer **126L** such that portions of the substrate **102**, portions of the active extension regions **122**, **124**, a portion of the patterned control gate dielectric layer **114** and the top surface of smaller length control gate electrode **116a** may be exposed.

(36) Referring to FIG. **10**, a second ion implantation step **128** may be performed. In this step, the exposed portions of the active extension regions **122**, **124** may be ion implanted to form deep active regions **130**, **132** adjacent the active extension regions **122**, **124**. The deep active regions **130**, **132** may have an implanted ion concentration in the range of 1×10^{20} to 5×10^{21} , such as 2×10^{20} to 2×10^{21} , although lesser or greater dopant concentrations are within the contemplated scope of disclosure. The ions implanted in the second ion implantation step **128** may be the same as or different from the ions implanted in the first ion implantation step **120**. Together, one of the active extension regions **122**, **124** and the adjacent deep active region **130**, **132** form a source region and the other of the active extension regions **122**, **124** and the adjacent deep active region **130**, **132** for a drain region.

(37) Referring to FIG. **11**, a silicide layer **134** may be optionally formed on the top surfaces of the deep active regions **130**, **132**. The silicide layer **134** may provide enhanced electrical connection with subsequently formed via contacts. A photoresist layer (not shown) may be deposited over the surface of the intermediate structure illustrated in FIG. **10** and patterned to expose top surfaces of the deep active regions **130**, **132**. The silicide layer **134** may then be formed by depositing a thin layer of metal (not shown) and heating to react the metal with the deep active regions **130**, **132**. Further, as illustrated, if the smaller length control gate electrode **116a** is made of polysilicon, metal may be deposited on the top surface of the smaller length control gate electrode **116a** and heated for form a silicide layer **134** on the top surface of the control gate electrode **116a**.

(38) Referring to FIG. **12**, an interlevel dielectric layer **136** may be deposited over the entirety of

the intermediate structure illustrated in FIG. 11. Via holes (not shown) may be formed in the interlevel dielectric layer **136** and subsequently filled with metal, such as W, Cu, Co, Mo, Ru, other elemental metals, or an alloy or a combination thereof. In this manner, electrical contacts **138** to the deep active regions **130**, **132** may be formed to complete a flash memory cell **100**. The formation of the interlevel dielectric layer **136** and the electrical contacts **138** completes a flash memory cell **100**. In an embodiment, the flash memory cell **100** may have a first electrical contact **138** electrically connected to one of the deep active regions **130**, **132** (source or drain) through optional silicide layer **134**, a second electrical contact **138** electrically connected to the other deep active region **130**, **132** (source or drain) through optional silicide layer **134**, and a third electrical contact **139** electrically connected to the smaller length control gate electrode **116a** through optional silicide layer **134**. A fourth electrical contact **140a** may be electrically connected to the patterned control gate dielectric layer **114** through optional top layer of the patterned second oxide layer **112**. In an embodiment, the flash memory cell **100** may include a fifth electrical contact **140b** electrically connected to the patterned control gate dielectric layer **114** through optional top layer of the patterned second oxide layer **112**. In various embodiments, the flash memory cell **100** may be configured such that the flash memory cell **100** may be programmed and erased by applying a voltage to the fourth electrical contact **140a** and read by applying a voltage to the fifth electrical contact **140b**.

(39) FIG. 13 illustrates a two dimensional array **200** of flash memory cells **100**. To electrically isolate columns of adjacent flash memory cells **100**, shallow trench isolation (STI) structures **150** may be formed in the semiconductor substrate **102**. The shallow trench isolation structures **150** may be formed by etching trenches (not shown) in the semiconductor substrate **102** and filling the trenches with a dielectric material, such as SiO₂. The STI trenches may have a depth in a range from 50 nm to 500 nm, such as 100 nm to 400 nm, although lesser or greater depths are within the contemplated scope of disclosure. For example, the STI trenches **150** may have a depth in a range of 75 nm to 400 nm, 100 nm to 350 nm, although lesser or greater depths are within the contemplated scope of disclosure.

(40) In various embodiments, the two dimensional array **200** of flash memory cells **100** includes a two-dimensional periodic array of floating gate electrodes **106** and smaller length control gate electrode **116a** located over a semiconductor substrate **102** which has a doping of a first conductivity type. The smaller length control gate electrode **116a** may be located over the floating gate electrodes **106**. The length L_{sub.CG} along a major axis of the smaller length control gate electrode **116a** may be less than a length L_{sub.FG} along a major axis of the floating gate electrode **106**. The two dimensional array **200** of flash memory cells **100** also includes a two-dimensional array of deep active regions **130**, **132** that may be formed within the semiconductor substrate **102**. The deep active regions **130**, **132** have a doping of a second conductivity type. The two-dimensional array of deep active regions **130**, **132** may be laterally offset from the two-dimensional array of floating gate electrodes **106** along a first horizontal direction. Each of the floating gate electrodes **106** may be located between a neighboring pair of deep active regions **130**, **132** within the two-dimensional array of deep active regions **130**, **132**. In various embodiments, each of the two-dimensional array of floating gate electrodes **106**, the two-dimensional array of deep active regions **130**, **132**, and the two-dimensional array of smaller length control gate electrode **116a** has a same first pitch along the horizontal direction.

(41) FIG. 14 is a flowchart illustrating general method **300** of making a memory device according to various embodiments. Referring to step **302**, a continuous tunnel dielectric layer **104L** may be deposited over a semiconductor substrate **102**. The tunnel dielectric may be made of any suitable dielectric material and may have a thickness in the range of 1-15 nm. Referring to step **304** a continuous floating gate layer **106L** may be deposited over the continuous tunnel dielectric layer **104L**. Referring to step **306**, a continuous control gate dielectric layer **114L** may be deposited over the continuous floating gate layer **106L**. In various embodiments, the continuous control gate

dielectric layer **114L** may comprise a continuous first oxide layer **108L**, a continuous nitride layer **110L** and a continuous second oxide layer **112L**. Referring to step **308**, a continuous control gate layer **116L** may be deposited over the continuous control gate dielectric layer **114L**. The continuous tunnel dielectric layer **104L**, continuous floating gate layer **106L**, continuous control gate dielectric layer **114L** and continuous control gate layer **116L** may be deposited by any suitable method such as CVD, OMCVD, PECVD, LPCVD or ALD. Referring to step **310**, the continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L** and the continuous control gate layer **116L** may be patterned to form a patterned tunnel dielectric layer **104**, a floating gate electrode **106**, a patterned control gate dielectric layer **114** and a control gate electrode **116**. Patterning may be performed by depositing a continuous photoresist layer **118L** over the continuous control gate layer **116L** and using photolithography to pattern an appropriate mask over the continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L** and the continuous control gate layer **116L**. The exposed layers of the continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L** and the continuous control gate layer **116L** may be etched to form the control gate electrode **116**, patterned control gate dielectric layer **114**, floating gate electrode **106** and patterned tunnel dielectric layer **104**. Any suitable etching method such as wet etching, dry etching or their combination may be used to form the control gate electrode **116**, the patterned control gate dielectric layer **114**, the floating gate electrode **106** and the patterned tunnel dielectric layer **104**. Referring to step **312**, the control gate electrode **116** may be further patterned to form a smaller length control gate electrode **116a**. The smaller length control gate electrode **116a** may have a length $L_{\text{sub.CG}}$ parallel to the major axis of the semiconductor substrate **102** that is shorter than a length $L_{\text{sub.FG}}$ of the floating gate electrode **106**.

(42) Generally, the structures and methods of the present disclosure may be used to form flash memory cells **100** in which at least some of the flash memory cells **100** have a smaller length control gate electrode **116a** which is shorter in length $L_{\text{sub.CG}}$ than the floating gate electrode **106**. Further, at least some of the flash memory cells **100** may have at least one electrical contact **140a**, **140b** to the patterned control gate dielectric layer **114** in addition to the electrical contacts **138** to the deep active regions **130**, **132** (source/drain regions) and the third electrical contact **139** to the smaller length control gate electrode **116a**. The floating gate electrode **106** may be programmed and erased by applying a voltage to the electrical contact **140a**, **140b** connected to the patterned control gate dielectric layer **114**. By using the electrical contact **140a**, **140b** to the patterned control gate dielectric layer **114** for programming and erasing rather than using the smaller length control gate electrode **116a**, the resistance $R_{\text{sub.poly}}$ associated with the (polysilicon) smaller length control gate electrode **116a** may be avoided. The elimination of the resistance $R_{\text{sub.poly}}$ associated with the polysilicon smaller length control gate electrode **116a** may result in a faster memory device. In some embodiments, more than one electrical contact **140a**, **140b** may be provided to the control gate dielectric layer **114**. This may result in an increased coupling area which may lead to a higher coupling ratio. The increase in the coupling ratio may also result in a faster flash memory cell **100**.

(43) An embodiment is drawn to a flash memory cell **100** located on a semiconductor substrate **102** including a floating gate electrode **106**, a patterned tunnel dielectric layer **104** located between the semiconductor substrate **102** and the floating gate electrode **106**, a smaller length control gate electrode **116a** and a patterned control gate dielectric layer **114** located between the floating gate electrode **106** and the smaller length control gate electrode **116a**. The length $L_{\text{sub.CG}}$ of a major axis of the smaller length control gate electrode **116a** is less than a length $L_{\text{sub.FG}}$ of a major axis of the floating gate electrode **106**.

(44) An embodiment is drawn to a two dimensional array **200** of flash memory cells **100** including a two-dimensional array of floating gate electrodes **106** and smaller length control gate electrode

116a located over a semiconductor substrate **102** having a doping of a first conductivity type. The smaller length control gate electrode **116a** may be located over the floating gate electrodes **106** and have a length $L_{sub.CG}$ along a major axis of the smaller length control gate electrode **116a** that is less than a length $L_{sub.FG}$ along a major axis of the floating gate electrode **106**. The two dimensional array **200** also includes a two-dimensional array of deep active regions **130**, **132** that may be formed within the semiconductor substrate **102**. The deep active regions **130**, **132** may have a doping of a second conductivity type. Also, the deep active regions **130**, **132** may be laterally offset from the two-dimensional array of floating gate electrodes **106** along a first horizontal direction. Each of the floating gate electrodes **106** may be located between a neighboring pair of deep active regions **130**, **132** within the two-dimensional array of deep active regions **130**, **132**. (45) In various embodiments, each of the two-dimensional array of floating gate electrodes **106**, the two-dimensional array of deep active regions **130**, **132**, and the two-dimensional array of smaller length control gate electrode **116a** may have a same first pitch $P_{sub.1}$ along the first horizontal direction. In various embodiments, the array of flash memory cells **100** comprises a two-dimensional periodic array of flash memory cells **100**. Each flash memory cell **100** in the two-dimensional periodic array comprises a floating gate electrode **106** in the two-dimensional array of floating gate electrodes **106**, deep active regions **130**, **132** in a two-dimensional array of deep active regions **130**, **132**, and a smaller length control gate electrode **116a** in the two-dimensional array of smaller length control gate electrode **116a**. Each flash memory cell **100** in the two-dimensional periodic array **200** may be laterally offset from an adjacent flash memory cell **100** in the two-dimensional periodic array **200** along a second horizontal direction with a same second pitch $P_{sub.2}$.

(46) An embodiment is drawn to a method **300** of making a flash memory device **100** including depositing a continuous tunnel dielectric layer **104L** over a semiconductor substrate **102**, depositing a continuous floating gate layer **106L** over the continuous tunnel dielectric layer **104L**, depositing a continuous control gate dielectric layer **114L** over the continuous floating gate layer **106L**, depositing a continuous control gate layer **116L** over the continuous control gate dielectric layer **114L**, patterning the continuous tunnel dielectric layer **104L**, the continuous floating gate layer **106L**, the continuous control gate dielectric layer **114L** and the continuous control gate layer **116L** to form a patterned tunnel dielectric layer **104**, a floating gate electrode **106**, a patterned control gate dielectric layer **114** and a control gate electrode **116** and further patterning the control gate electrode **116** to form a smaller length control gate electrode **116a** such that the smaller length control gate electrode **116a** has a length $L_{sub.CG}$ parallel to the major axis of the substrate **102** that is shorter than a length $L_{sub.FG}$ of a major axis of the floating gate electrode **106**.

(47) In an embodiment, active extension regions **122**, **124** may be formed in the semiconductor substrate **102** extending laterally from sidewalls of the patterned tunnel dielectric layer **104**. The active extension regions **122**, **124** may be formed by a first ion implantation step **120**. In an embodiment, deep active regions **130**, **132** may be formed with a second ion implantation step **128**.

(48) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method of forming a memory cell, the method comprising: forming a gate stack on a substrate, wherein the gate stack includes a control gate dielectric layer and a control gate layer on the control gate dielectric layer; etching the control gate layer to reduce a length of the control gate layer in a first direction and expose an upper surface of the control gate dielectric layer; forming a spacer structure on the gate stack and having a frame-shaped cross-section, wherein the spacer structure comprises: a first sidewall spacer on a sidewall of the control gate dielectric layer; and a second sidewall spacer on a sidewall of the control gate layer and on the upper surface of the control gate dielectric layer; forming a source region and a drain region in the substrate, such that a channel region is located beneath the gate stack in the substrate; and forming an electrical contact on the upper surface of the control gate dielectric layer over the channel region.
2. The method of claim 1, wherein the forming of the spacer structure comprises: after the etching of the control gate layer, depositing a layer of spacer dielectric material on the gate stack and on a surface of the substrate; and etching the layer of spacer dielectric material to form the first sidewall spacer and the second sidewall spacer.
3. The method of claim 2, wherein the etching of the layer of spacer dielectric material comprises etching a portion of the layer of spacer dielectric material on the upper surface of the control gate dielectric layer to form the second sidewall spacer.
4. The method of claim 1, wherein the forming of the spacer structure comprises forming the spacer structure continuously around the gate stack.
5. The method of claim 1, further comprising: forming a patterned photoresist layer on the gate stack, wherein the etching of the control gate layer comprises etching the control gate layer through an opening in the photoresist layer.
6. The method of claim 1, wherein the forming of the gate stack comprises: forming a tunnel dielectric layer on the substrate; forming a floating gate layer on the tunnel dielectric layer; and forming the control gate dielectric layer on the floating gate layer.
7. The method of claim 6, wherein the forming of the spacer structure comprises forming the first sidewall spacer on a sidewall of the tunnel dielectric layer and on a sidewall of the floating gate electrode.
8. The method of claim 1, wherein the first sidewall spacer has a height that is greater than a height of the second sidewall spacer on the upper surface of the control gate dielectric layer.
9. The method of claim 1, wherein an upper surface of the first sidewall spacer is substantially coplanar with the upper surface of the control gate dielectric layer.
10. The method of claim 1, wherein forming of the electrical contact comprises forming the electrical contact in a central region of the exposed upper surface of the control gate dielectric layer.
11. The method of claim 1, wherein the forming of the gate stack comprises forming the control gate dielectric layer to include a first oxide layer on the floating gate electrode, a nitride layer on the first oxide layer, and a second oxide layer on the nitride layer.
12. The method of claim 11, wherein the forming of the control gate dielectric layer comprises forming the first oxide layer to have a first thickness and forming the second oxide layer to have a second thickness greater than the first thickness.
13. The method of claim 1, further comprising: forming a pair of isolation regions contacting the source region and the drain region on opposing sides of the gate stack.
14. A method of forming a memory cell array, the method comprising: forming a plurality of isolation regions in a substrate; forming a plurality of first conductivity type regions in the substrate to be alternately formed with the plurality of isolation regions in a first direction; and forming a plurality of memory cells in the plurality of first conductivity type regions, wherein each memory cell of the plurality of memory cells includes: a gate stack comprising: a control gate dielectric layer having a first length in the first direction; and a control gate on the control gate dielectric

layer and having a second length in the first direction less than the first length; a spacer structure on the gate stack and having a frame-shaped cross-section, wherein the spacer structure comprises: a first sidewall spacer on a sidewall of the control gate dielectric layer; and a second sidewall spacer on a sidewall of the control gate layer and on the upper surface of the control gate dielectric layer; a source region and a drain region in the substrate and a channel region located beneath the gate stack in the substrate; and an electrical contact on the upper surface of the control gate dielectric layer over the channel region.

15. The method of claim 14, wherein the forming of the plurality of memory cells comprises forming the spacer structure continuously around the gate stack.

16. The method of claim 14, wherein the forming of the plurality of memory cells comprises forming the second sidewall spacer to have a width in the first direction substantially the same as a width of the first sidewall spacer in the first direction.

17. The method of claim 14, wherein the forming of the plurality of memory cells comprises forming the plurality of memory cells to have a first pitch in the first direction and a second pitch less than the first pitch in a second direction perpendicular to the first direction.

18. A method of forming a memory cell, the method comprising: forming a gate structure on a substrate, wherein the gate structure includes a control gate dielectric layer and a control gate layer on the control gate dielectric layer; forming extension regions of source and drain regions in the substrate on opposite sides of the gate structure in a first direction; after the forming of the extension regions, etching the control gate layer to reduce a length of the control gate layer in the first direction and expose an upper surface of the control gate dielectric layer; forming a spacer structure on the gate structure; after the forming of the spacer structure, forming deep regions of the source and drain regions in the substrate on opposite sides of the gate structure in the first direction, such that a channel region is located beneath the gate structure in the substrate; and forming an electrical contact on the upper surface of the control gate dielectric layer over the channel region.

19. The method of claim 18, wherein the forming of the spacer structure comprises: forming a first sidewall spacer on a sidewall of the control gate dielectric layer; and forming a second sidewall spacer on a sidewall of the control gate layer and on the upper surface of the control gate dielectric layer.

20. The method of claim 18, further comprising: forming an interlevel dielectric layer on the gate structure such that the interlevel dielectric layer contacts the control gate dielectric layer between the electrical contact and the second sidewall spacer.
